

Aditya Raj

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EDUCATION

Indian Institute of Engineering Science and Technology
B.Tech. Information Technology

August 2027
Current CGPA: 8.51/10.0

SKILLS

Languages: C/C++, JavaScript/TypeScript, HTML/CSS, Python

Technologies/Tools: Git, GitHub, MongoDB, Postgresql, VectorDB, Websockets, JWT, OAuth, GraphQL

Frameworks/Libraries: React, Next.js, Express, Socket.IO, Recoil, Redux, Zustand, Mongoose, TailwindCSS, Prisma

WORK EXPERIENCE

NewsNest — AI news startup | *Founding Engineer* 25th May 2025 - 20th July 2025
Built an AI-driven news aggregator that scrapes website RSS, YouTube videos/Shorts, and Twitter (via **Puppeteer**), scheduled via **GitHub Actions**. Used **Pinecone** and **Groq LLaMA** inference for generating **RAG** based Timeline and categorization of news. Designed and coded complete Backend using **GraphQL, Express, TypeScript, Prisma, PostgreSQL**

PhotoGPT — AI Image Generation | *Software Engineer* 28th July 2025 - current
Built UI for React image editor using **Fabric.js** to upload, smudge, and mask regions for AI-based face-swap. Integrated with backend to send marked areas for **inpainting** and regeneration.

PROJECTS

Tubespace | *Prisma, Typescript, Socket.IO, Zustand, Express, React, YouTube APIs*

GitHub | Live Link March, 2025

- Built a YouTube publishing platform with server-side uploads to bypass user bandwidth reliability.
- Integrated **OAuth 2.0** with **YouTube Data API** for secure uploads without credential sharing.
- Implemented chunked video uploads to YouTube by streaming from Cloudinary using **got**, with real-time progress relayed via secure WebSocket, powered by **BullMQ + worker architecture**.
- Designed a multi-stage upload pipeline (video → thumbnail → metadata) as **queued tasks** triggered post **OAuth** authentication and session URL generation.
- Enabled video preview via **Cloudinary** and one-click approval to publish videos with metadata, thumbnails, tags, and privacy settings.
- Built a responsive React frontend using **TypeScript, Zustand** for global state, **React Hook Form** for type-safe forms, and **React Query** for reliable API integration.
- Integrated **Resend API** and **Firebase Messaging** for real-time email and push notifications.
- Modeled users, chats, and tasks using **PostgreSQL + Prisma ORM**, with secure auth via **JWT** and **Redis-backed OTP system**.

Midfruits | *Socket.IO, Node.js, React, Recoil, framer-motion*

GitHub | Live Link October, 2024

- Built a real-time **multiplayer web game** enabling players to create custom rooms, invite friends, and compete in dynamic fruit-catching battles.
- Implemented fast **bi-directional communication** with **Socket.IO**, handling live movement sync, scores, collision detection, and fruit generation.
- Used **Recoil** for managing complex player/UI state and **framer-motion** for smooth animations.
- Won **1st place** in a college game dev competition; gained **1000+ page views** within two weeks of launch.

ACHIEVEMENTS

WebGame Challenge Winner | 1st place in college WebGame challenge

Hacktoberfest' 2024 and GSSOC'2023 | Contributed using CSS and JavaScript.